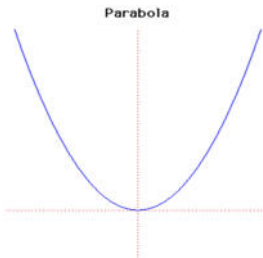
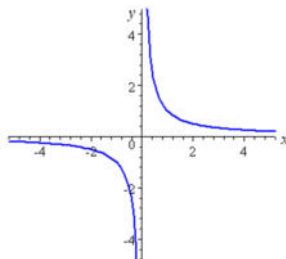


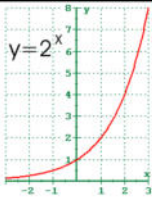
TOPIC - FUNCTIONS

SKILLS NEEDED BEFORE ATTEMPTING THIS SECTION

1.	Function Notation
2.	Understanding and sketching of 3 basic functions • Quadratic function (Parabola) • Hyperbolic function (Hyperbola) • Exponential function
3.	Sketching of all 3 functions involving a vertical shift ('q')
4.	Sketching of all 3 functions involving the changes that 'a' makes.
5.	Finding domain and range of any function.

GLOSSARY OF TERMS

Function	A relation for which each element of the domain (x) corresponds to exactly one element of the range (y). For every x-value there is only one possible y-value. To test for a function on a graph, use the 'horizontal line test' – run a ruler from left to right. If your ruler only ever touches the graph in one spot, it is a function.
Parabola	A function of the form $y = ax^2 + bx + c$ or $y = a(x - p)^2 + q$ 
Hyperbola	A function of the form $y = \frac{a}{x-p} + q$ 
Exponential	A function of the form $y = ab^{x-p} + q$

	
Increasing function	A function that is going 'uphill' when looking at it from left to right. If a function has a turning point, this is the point at which a function could change from increasing to decreasing.
Decreasing function	A function that is going 'downhill' when looking at it from left to right. If a function has a turning point, this is the point at which a function could change from decreasing to increasing.
Average Gradient	The gradient between two points on a curved graph.
Domain	All the possible x-values that are covered by the function/graph. (To find the domain always study the graph from left to right)
Range	All the possible y-values that are covered by the function/graph. (To find the range always study the graph from bottom to top)
x-intercept	Where a function crosses (intercepts) the x-axis
y-intercept	Where a function crosses (intercepts) the y-axis
Axis of symmetry	A line that cuts the function exactly in half. A parabola has a vertical line of symmetry ($y = \dots$) and a hyperbola has 2 axes of symmetry (which form a cross $\rightarrow X$. These are in the form $y = mx + c$) An exponential graph does not have an axis of symmetry.
Asymptote	A straight line that a curved graph gets closer and closer to but never touches it.
Turning Point	The point at which a function (parabola, sin or cos graph) changes from increasing to decreasing OR from decreasing to increasing.
Maximum	Only involves functions that have a turning point. The maximum is the highest the graph can be and is always the y-value of the turning point.
Minimum	Only involves functions that have a turning point. The minimum is the lowest the graph can be and is always the y-value of the turning point.

SUMMARY OF KEY CONCEPTS

Steps to sketching a parabola ($y = ax^2 + bx + c$):

1. Shape (If $a > 0$, it 'smiles' (convex); If $a < 0$, it is 'sad' (concave))
2. y-Intercept (Let $x = 0$ and solve for y)
3. x-Intercepts (Let $y = 0$ and solve for x)
4. Axis of Symmetry (Ensure equation is in standard form: $y = ax^2 + bx + c$ and use $x = \frac{-b}{2a}$)
5. Turning Point (Substitute value of x from Axis of symmetry and find corresponding y co-ordinate)

Finding equations of parabolas

1. Finding the equation of a parabola given the TURNING POINT and another point.

Use: $y = a(x - p)^2 + q$	Example: The turning point of a parabola is (2 ; 6) and it also passes through the point (5 ; -30). Find the equation of the parabola.
Step 1: Substitute turning point into (p; q)	$y = a(x - 2)^2 + 6$
Step 2: Substitute other point (x ; y)	$-30 = a(5 - 2)^2 + 6$
Step 3: Solve for a	$-30 = a(9) + 6$ $-36 = 9a$ $-4 = a$
Step 4: Substitute this back into (1)	$y = -4(x - 2)^2 + 6$
Step 5: Multiply out and collect like terms ($y = ax^2 + bx + c$)	$y = -4(x^2 - 4x + 4) + 6$ $y = -4x^2 + 16x - 16 + 6$ $y = -4x^2 + 16x - 10$

2. Finding the equation of a parabola given the X-INTERCEPTS and another point.

Use: $y = a(x - x_1)(x - x_2)$	Example: A parabola passes through the point (-2 ; 0), (5 ; 0) and (0 ; 5). Find the equation of the graph.
Step 1: Substitute 2 values of x-intercepts into x_1 and x_2	$y = a(x + 2)(x - 5)$
Step 2: Multiply out (1)	$y = a(x^2 - 3x - 10)$ (1)
Step 3: Substitute the other co-ordinate (x ; y) and solve for a.	$5 = a(-10)$ $\frac{-1}{2} = a$
Step 4: Substitute 'a' back into (1)	$y = \frac{-1}{2}(x^2 - 3x - 10)$
Step 5: Multiply out and collect like terms ($y = ax^2 + bx + c$)	$y = \frac{-1}{2}x^2 + \frac{3}{2}x + 5$

Minimum and maximum values of quadratic expressions (parabolas)

Once you have completed the square of an expression and have it in the form:

$$a(x - p)^2 + q$$

If $a > 0$, there will be a
MINIMUM value

If $a < 0$, there will be a
MAXIMUM value

This will be the MAXIMUM
or MINIMUM value

Example 1:

Has a maximum value $\swarrow$ $-2(x - 1)^2 + 4$
which is $\nearrow$

Example 2:

Has a minimum value $\swarrow$ $\frac{1}{2}(x - 5)^2 - 5$
which is $\nearrow$

Steps to sketching a hyperbola ($y = \frac{a}{x-p} + q$):

1. Decide which quadrants it lies in (If $a > 0$, quadrants 1 & 3; If $a < 0$, quadrants 2 & 4)
2. Find equations of asymptotes. (From $y = \frac{a}{x-p} + q$, asymptotes will be $x = p$ which is the horizontal asymptote and $y = q$ which is the vertical asymptote.
3. y-intercept (Let $x = 0$ and solve for y)
4. x-intercepts (Let $y = 0$ and solve for x)
5. It is recommended to find at least one more point. Substitute any x -value and find the corresponding y -value.

Finding the equation of a hyperbola

Needed: The horizontal and vertical asymptotes and at least one other point on the graph.

Start with writing down the equation: $y = \frac{a}{x-p} + q$

1. Use the horizontal asymptote to put in the value of 'q'
2. Use the vertical asymptote to put in the value of 'p'
3. Substitute the other point into x and y's place to solve for 'a'

Steps to sketching an exponential graph ($y = ab^{x-p} + q$)

1. Decide whether it lies above the x-axis or below. (If $a > 0$, it lies above the x-axis and if $a < 0$ it lies below the x-axis) However, if there is a vertical shift it will not remain completely above or below.
2. Decide whether it is increasing or decreasing. (If $b > 1$, it will increase. In other words it will go up from left to right. If $0 < b < 1$, it will decrease. In other words it will go down from left to right)
3. Find the equation of the asymptote. ($y = q$)
4. y-intercept (Let $x = 0$ and solve for y)
5. x-intercepts (Let $y = 0$ and solve for x)

Finding the equation of an exponential graph:

The amount of information required is always directly related to the number of variables missing. You are likely to be given the 'format' of the graph. For example, $y = ab^x + q$ (notice there is no 'p'). As you are looking for the values of 3 variables (a, b and q) there must be 3 pieces of information given to you.

1. Use the horizontal asymptote to put in the value of 'q'
2. Substitute any other information given to find other missing values. You may have to use simultaneous equations if there are two points given and two variables to find.

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